KANNAN APPIYA SANTHARAM

Senior Lead Software Engineer (Frontend/Full Stack) | React | TypeScript | Node.js | AI-Native
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PROFESSIONAL SUMMARY

10.5+ years in enterprise SaaS, building across React, TypeScript and Node.js. Currently leading a frontend team on a platform managing millions of customer endpoints, owning architecture across a 12-package monorepo, design systems and performance, and automating most frontend work with AI agents, cutting new page development from 2 weeks to about 5 hours. Keen to bring strong frontend and full stack skills and team leadership to a new role.

KEY ACHIEVEMENTS

- **Automated most frontend work** with AI agents that turn designs into production code, integrate APIs and write tests, cutting new page development from 2 weeks to about 5 hours.
- **Lead a frontend team of 6** at SuperOps on a platform managing millions of endpoints, shipping weekly and owning hiring, performance reviews, architecture and the roadmap with product.
- **Enabled 50+ enterprise accounts** at Freshworks, $100K to $1M ARR each, with custom Node.js middleware and microservices that met requirements the core product could not.
- **Delivered custom features and public apps** across 4 Freshworks products to 10,000+ customers, built in React with Node.js services on AWS, secured with JWT and, for public apps, OAuth 2.0 where the third-party API supported it.
- **Architected an AI-driven end-to-end testing platform** that cut full product regression from 6 hours of manual testing to a 40-minute automated run.
- **Cut product initial load time by 70%**, from 5 seconds to 1.5 seconds.
- **Hired 9 engineers** and mentored 10+ across SuperOps and Freshworks, at 90% retention.

SKILLS

Frontend & Full Stack: React, Next.js, TypeScript, Node.js, Python, Microservices, GraphQL, REST API Design, JWT & OAuth 2.0, MySQL, DynamoDB, Tailwind CSS, Playwright, Jest & Vitest

Platform & Performance: Monorepos (Turborepo, pnpm), Micro-Frontends & Module Federation, Vite, Rspack & Webpack 5, Design Systems, Core Web Vitals, Jenkins CI/CD, Docker, AWS

AI-Native Engineering: LLM Orchestration, Model Context Protocol (MCP) Servers, Agent Skills & Guardrails, RAG & Embeddings, Amazon Bedrock AgentCore, Claude Code

Leadership: Team Leadership (6 engineers), Hiring & Mentorship (10+ engineers), Solution Architecture & System Design, Cross Team Collaboration, Stakeholder & Customer Management

WORK EXPERIENCE

SUPEROPS

Chennai, India

Senior Lead Software Engineer

Jul 2022 – Present

Progression: Senior Software Engineer to *Lead Software Engineer* to *Senior Lead Software Engineer*

- Automated most of our frontend work with AI agents that turn designs into production code, integrate APIs from backend contracts and write Playwright tests behind strict commit checks, cutting new page development from 2 weeks to about 5 hours, on a new frontend monorepo set up from scratch on Vite, Turborepo and pnpm.
- Hands-on delivery of product features in React and TypeScript, including role based access control (RBAC) and near real time operational dashboards using cached Apollo GraphQL hooks that balance data freshness against network load, with product and UI/UX teams.
- Manage the team's sprint planning, architecture reviews, interviews, mentoring and performance feedback, and roadmap prioritisation with product managers.

- Reduced dev build time by 96%, from 2 minutes to 5 seconds, through a solo Webpack 5 to Rspack migration delivered in 3 weeks across 12 monorepo packages, reclaiming roughly 50 engineer-hours a month across a 15-engineer frontend org.
- Architected an AI testing platform in React and Node.js where LLM agents write, run and fix a 300+ case Playwright suite and add tests for each new feature, gating every release in Jenkins CI/CD.
- Improved developer productivity with custom Claude Code skills and MCP servers in the production workflow, behind tiered write limits so engineers can run autonomous agents without putting core platform interfaces at risk.
- Built a Python MCP server exposing the frontend codebase as a queryable graph, packaged as a Docker image and hosted on AWS EC2 behind token-based authentication, now the shared context source for 3 engineering squads' AI tooling.
- Built a headless Base UI design system of 30+ reusable components now used across 40+ pages and sections, with customised design tokens kept in sync between Figma and code.
- Cut the product's initial page load from 5 seconds to 1.5 seconds through route-level code splitting, React lazy loading and per-package bundle chunking, ending the slowness-related support tickets, and removed 30,000+ lines of dead code and unused dependencies.

FRESHWORKS *Chennai, India*
Senior Software Engineer (Customer-Facing Engineering) *Jun 2018 – Jul 2022*
Progression: Onboarding Engineer to *Senior Software Engineer*

- Technical point of contact for 100+ mid-market and enterprise accounts over four years, partnering with Support, Customer Success and Sales to turn account requirements into shipped solutions, recognised with the Customer Champion Award.
- Built tier-1 SaaS integrations over REST API services, including the Jira integration adopted by 4,000+ Freshworks customers, alongside Salesforce, Zendesk and ServiceNow integrations for those accounts, expanding platform fit in enterprise deals.
- Shipped custom features and public apps in React and Node.js across 4 Freshworks products to 10,000+ customers, with JWT for app authentication and OAuth 2.0 where the third-party API supported it.
- Built the Node.js middleware and microservices behind 50+ enterprise accounts worth $100K to $1M ARR each, deployed on AWS EC2 with S3, DynamoDB and MySQL, delivering requirements the core product could not and keeping onboarding and renewals on track.
- Mentored 7 junior and onboarding engineers on React, REST API development and UI architecture through structured code and design reviews.

NICHE VIDEO MEDIA / INFIGENIC *Chennai & Bengaluru, India*
Software Developer (acquired by Freshworks) *Mar 2016 – Jun 2018*
Progression: Web Application Developer & Designer to *Software Developer*

- Built an admin dashboard that listed the features, bandwidth and storage limits behind each Stripe plan and let admins create custom plans on demand, increasing self-serve revenue.
- Built video player customisation from scratch with configurable controls, annotations and call-to-action overlays, increasing viewer conversion.
- Built responsive, high-converting company and product landing pages.
- Integrated Freshservice with DocuSign over REST API services, unlocking a $500K ARR enterprise deal, and joined Freshworks with the team on acquisition.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science *2011 – 2015*
K.L.N. College of Information Technology (Affiliated with Anna University), Tamil Nadu, India

LANGUAGES

English: Fluent (Professional Working Proficiency) | **Tamil:** Native | **Hindi:** Intermediate